

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460777.8459 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.055
Transit depth (Rp/Rs)^2	3.24 +/- 1.96 [%]
Orbital Inclination (inc)	89.37 +/- 1.73
Airmass coefficient 1 (a1)	0.0413 +/- 0.0084
Airmass coefficient 2 (a2)	2.92 +/- 1.67
Scatter in the residuals of the lightcurve fit is	19.72 %
Best Comparison Star	#1 - [420, 475]
Optimal Aperture	22.04
Optimal Annulus	205.71
Transit Duration (day)	0.093 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.055 vs 0.1406 (deviation 0.72σ)
Duration fit vs published	0.093 d vs 0.0975 d (deviation 0.45σ)
Mid-time O–C	11.8 min
Depth SNR	3.3
Residual scatter	19.72 %

Light curve

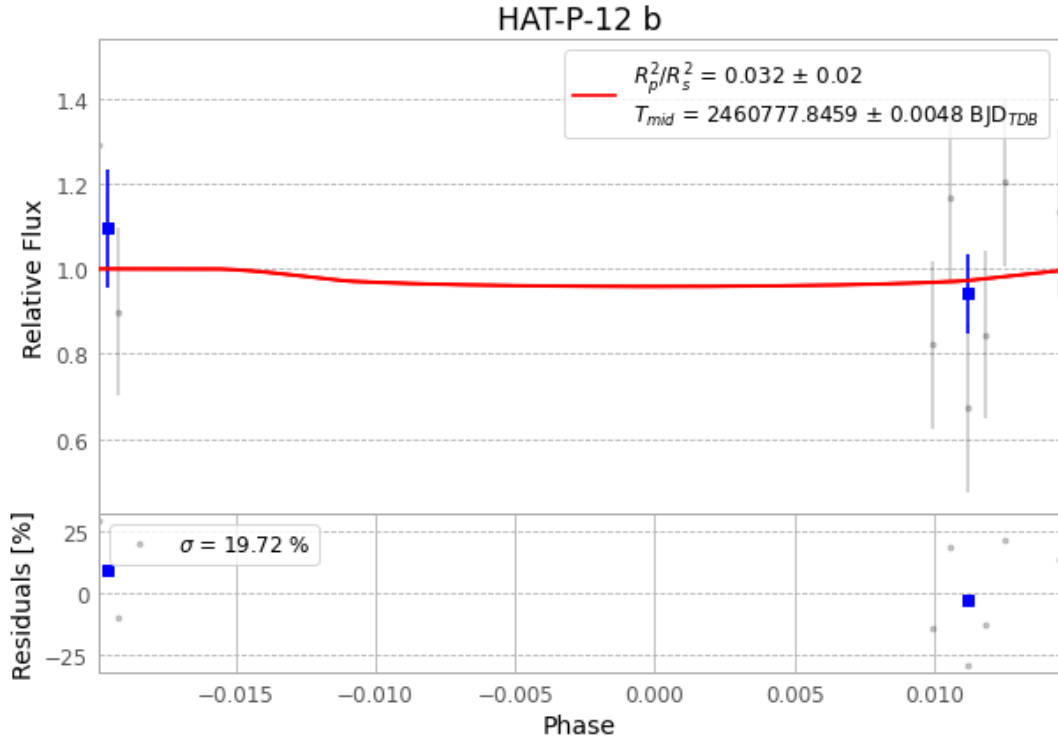


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-04-11.png (SHA-256 68c18edd88313766...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 478], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 90e677845c4a81fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), HATP-12250412063915.FITS → HATP-12250412101216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250412.log (a72cfe565902...), FinalLightCurve_HAT-P-12 b_2025-04-11.png (68c18edd8831...), FinalLightCurve_HAT-P-12 b_2025-04-11.csv (c467b94e1471...)

Compute resources

Wall time 660.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:22Z.